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Reflection Journal

Reflection Journal Structure

Brian: Introduction + Personal Reflection

The focus of Lab 04 was on using Python to examine and analyze the data by going through the entire exploratory data analysis (EDA) process. During the lab, I used the Titanic dataset and carried out my study and produced visualizations with the help of tools such as Pandas, Matplotlib, and Seaborn. The aim of the lab was not just to write and run Python code, but also to understand what the data was revealing and how the various variables could be compared. I realized that data can be organized, investigated, and displayed in the form of graphs so that patterns become easier to understand. I also realized that each stage of the workflow is linked together since the variables in the code can influence the results and the visualizations that are obtained. As a result, I gained a better appreciation of how the lab can act as an important preliminary step before concluding a dataset.

The way I approached Lab 04 enabled me to gain a better understanding of how to work with data and Python code. One challenge I encountered was ensuring I used the right columns for each experiment and that the comments and graph titles accurately described what the code

was doing. For instance, I had to work out the difference between SibSp and Embarked since they provide diverse kinds of information in the Titanic dataset. I also needed to understand how x, y, and hue could influence a Seaborn visualization and how altering these values changes what the graph shows. Whenever I came across something that did not match the experiment, I went back to the code, checked the meaning of each variable, and made the necessary corrections rather than simply running the code. I have learned that while getting the code to run successfully is important, it is only one aspect of data analysis; it is equally important to ensure that the results are accurate and meaningful. As a result of this experience, I now feel more confident in reading Python code, interpreting the visualizations, and verifying my work before proceeding to the next section of the lab.

Toan: Machine learning Workflow + Personal reflection

In this lab, I worked through a machine learning workflow by using the dataset provided in the notebook and completing different coding and EDA activities. I used Python tools such as pandas, NumPy, and Matplotlib to work with the data, examine the dataset, and understand the information it contained. I checked the data, looked at different values and patterns, and created visualizations to make the results easier to understand. Running the code and seeing the graphs helped me understand how important data analysis is to learn workflow before building or evaluating a model.

One thing I learned from this lab is that a machine learning workflow is not just about writing code and getting an answer. Each step depends on the previous step, so it is important to understand the data before making conclusions from it. I also learned that visualizations could make patterns and relationships in the data easier to notice than looking only at numbers. When some code did not work as expected, I had to check the code and outputs and try again, which

helped me better understand what each step was doing. Overall, this lab gave me a better understanding of how data exploration, coding, and visualization work together in a machine learning project.

Montserrat: Improvements and Learning + Personal Reflection + Conclusion

One improvement I noticed during lab 04 was becoming more comfortable with interpreting data rather than focusing only on getting the Python code to run. Working with the Titanic dataset helped me understand how choosing different variables can change the information shown in a visualization. I also improved my ability to recognize patterns in graphs and troubleshoot problems when the results did not look correct. These skills will be useful in future machine learning and data-analysis projects because understanding the data, selecting appropriate features, and checking results carefully are important steps before building a model. Becoming more comfortable with Python and visualization tools will also make it easier to work with more complex datasets in future projects.

Conclusion

Overall, this lab showed me that data analysis is an important foundation for machine learning. I learned that creating visualizations is not just about making graphs look organized, but about using them to understand the information and recognize patterns in the data. My main takeaway from the lab is that taking time to examine and understand a dataset can lead to better analysis and more reliable results.